

HDOC DAY-41

Title

Menu-Driven C Program for String Operations Using Separate Functions

Aim

To write a **menu-driven C program** using separate user-defined functions to perform the following string operations:

1. Remove all vowels from a string.
 2. Find the first repeating lowercase alphabet in a string.
 3. Check whether two strings are anagrams of each other.
 4. Exit the program.
-

Algorithm

1. Remove All Vowels

1. Read a string from the user.
2. Traverse the string character by character.
3. Check whether each character is a vowel (a, e, i, o, u in uppercase or lowercase).
4. If the character is not a vowel, copy it to a temporary string.
5. Store the modified string back in the original string.
6. Display the string after removing vowels.

2. Find the First Repeating Lowercase Alphabet

1. Read a string from the user.
2. Create an array of size 26 and initialize all elements to zero.
3. Count the frequency of each lowercase alphabet.
4. Traverse the string again from the beginning.
5. The first lowercase character whose frequency is greater than 1 is the first repeating alphabet.
6. Display the character.
7. If no character repeats, display an appropriate message.

3. Check Whether Two Strings Are Anagrams

1. Read two strings from the user.

2. Compare their lengths.
 3. If the lengths are different, they cannot be anagrams.
 4. Count the frequency of characters in the first string.
 5. Subtract the frequency of characters appearing in the second string.
 6. If all frequency values become zero, the strings are anagrams.
 7. Otherwise, they are not anagrams.
-

C Program

```
#include <stdio.h>
#include <string.h>

// Function to remove all vowels
void removeVowels(char str[])
{
    int i, j = 0;
    char temp[200];

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] != 'a' && str[i] != 'e' &&
            str[i] != 'i' && str[i] != 'o' &&
            str[i] != 'u' && str[i] != 'A' &&
            str[i] != 'E' && str[i] != 'I' &&
            str[i] != 'O' && str[i] != 'U')
        {
            temp[j] = str[i];
            j++;
        }
    }

    temp[j] = '\0';
    strcpy(str, temp);
}

// Function to find first repeating lowercase alphabet
char firstRepeating(char str[])
{
    int count[26] = {0};
    int i;

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            count[str[i] - 'a']++;
        }
    }

    for (i = 0; str[i] != '\0'; i++)
    {
        if (str[i] >= 'a' && str[i] <= 'z')
```

```

        {
            if (count[str[i] - 'a'] > 1)
                return str[i];
        }

    return '\0';
}

// Function to check whether two strings are anagrams
int areAnagrams(char str1[], char str2[])
{
    int count[256] = {0};
    int i;

    if (strlen(str1) != strlen(str2))
        return 0;

    for (i = 0; str1[i] != '\0'; i++)
    {
        count[(unsigned char)str1[i]]++;
        count[(unsigned char)str2[i]]--;
    }

    for (i = 0; i < 256; i++)
    {
        if (count[i] != 0)
            return 0;
    }

    return 1;
}

int main()
{
    int choice;
    char str1[200], str2[200];
    char result;

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Remove all vowels from a string\n");
        printf("2. Find first repeating lowercase alphabet\n");
        printf("3. Check if two strings are anagrams\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a string: ");
                fgets(str1, sizeof(str1), stdin);
                str1[strcspn(str1, "\n")] = '\0';

```

```

        removeVowels(str1);

        printf("String after removing vowels: %s\n", str1);
        break;

    case 2:
        printf("Enter a string: ");
        fgets(str1, sizeof(str1), stdin);
        str1[strcspn(str1, "\n")] = '\0';

        result = firstRepeating(str1);

        if (result != '\0')
            printf("First repeating lowercase alphabet: %c\n",
result);
        else
            printf("No repeating lowercase alphabet found.\n");

        break;

    case 3:
        printf("Enter first string: ");
        fgets(str1, sizeof(str1), stdin);
        str1[strcspn(str1, "\n")] = '\0';

        printf("Enter second string: ");
        fgets(str2, sizeof(str2), stdin);
        str2[strcspn(str2, "\n")] = '\0';

        if (areAnagrams(str1, str2))
            printf("The strings are anagrams.\n");
        else
            printf("The strings are not anagrams.\n");

        break;

    case 4:
        printf("Program terminated.\n");
        break;

    default:
        printf("Invalid choice! Please try again.\n");
}

} while (choice != 4);

return 0;
}

```

Output

Output 1 – Remove All Vowels

```
===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit

Enter your choice: 1
Enter a string: Hello World
String after removing vowels: Hll Wrld
```

Output 2 – First Repeating Lowercase Alphabet

```
===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit

Enter your choice: 2
Enter a string: programming
First repeating lowercase alphabet: r
```

Output 3 – Check Anagrams

```
===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit

Enter your choice: 3
Enter first string: listen
Enter second string: silent
The strings are anagrams.
```

Output 4 – Exit

```
===== MENU =====
1. Remove all vowels from a string
2. Find first repeating lowercase alphabet
3. Check if two strings are anagrams
4. Exit

Enter your choice: 4
Program terminated.
```

Result

Thus, a **menu-driven C program using separate user-defined functions** was successfully written and executed to:

- Remove all vowels from a string.
- Find the first repeating lowercase alphabet.
- Check whether two strings are